UNIVERSITY OF PLYMOUTH
School of Engineering, Computing & Mathematics

COURSEWORK FEEDBACK

COMP3018 Human-Robot Interaction

Module Leader: **Dr. Amir ALY**

Academic Year: 2025/26

Assignment: **Set Exercises**
Student: **Alfie Nurse**
Mark: **81 %**

The general quality is indicated in the grid below before specific comments are provided. These comments will allow you to **benchmark your progress on this assignment** and they will indicate to you the parts that need improvement.

	+ 80%	70-79%	60-69%	50-59%	40-49%	30-39%	0-29%
Task (1) Cultural Diff. and HRI		√					
Task (2) Models for HRI	√						

Specific Comments

Task (1) Cultural Differences and HRI Design

- Strong and well-supported summary of Kaplan's East/West contrast, but it sometimes over-interprets Kaplan and drifts into broader HRI issues beyond the core summary asked.
- Thoughtful proposed factors for Africa with clear structure and relevant support, though some claims are quite broad across "Africa" and not all are equally well-grounded.
- Good translation of cultural factors into concrete appearance/behaviour traits for East, West, and Africa; this is one of the stronger sections, though a few design prescriptions are stated too confidently.
- Sensible adaptation of Kahn et al. to different regions, but it only develops a subset of the design patterns rather than engaging the pattern set more fully, so the breadth is limited.

Task (2) Models for Human-Robot Interaction

1. Strong answer that directly addresses the role of POMDPs in trust, cooperation, coordination, and collaboration, with a good conceptual distinction between these interaction modes; slightly held back by some overextension from lecture framing into the answer.
2. Clear and technically solid explanation of uncertainty, belief states, and Bayesian updating, showing good understanding of how POMDPs support decision-making under partial observability; it is more theoretical than concrete in parts, but still strong.
3. A very good discussion of trust modelling challenges, especially measurement, temporal dynamics, and computational difficulty, with relevant literature support; only a little more direct linkage to practical HRI examples was needed.
4. You develop a specific scenario, define state/action/observation/transition/reward components, and explain trust, uncertainty, benefits, and limitations in detail. It is impressive, though a little ambitious/speculative in places, especially around the OpenAI-based neuro-symbolic architecture.
5. A strong ethical discussion covering manipulation, explainability, privacy, data sovereignty, and latency-related social risk; well argued, though slightly narrower on wider social impacts beyond the chosen architecture.